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Modeling wildfires in the Alpine context: Lessons learnt from real case studies

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ABSTRACT

In the Alpine region, climate change is altering the predisposing factors of wildfires, leading to an increase in their frequency and intensity. This trend, combined with the inherent vulnerability of Alpine forest stands, underscores the growing importance of fire modeling as a fundamental tool to support both prevention strategies and emergency response. This study compares two wildfire simulation models — `FlamMap` and `PROPAGATOR` — which, although both widely used in Italy, are based on fundamentally different modeling approaches.

The models are applied to three significant case studies in Lombardia's Alpine area that took place in the period 2017–2022 (Tirano, Campolaro, and Sonico Wildfires), allowing a comparison of their performance in reproducing fire spread, rate of spread, and fireline intensity under varying topographic, vegetational, and meteorological conditions. Results indicate that both models reasonably captured the propagation of the three wildfires, although substantial differences emerged in fireline intensity and rate of spread, reflecting their distinct modeling assumptions. A closer examination of the outputs, combined with the specific input data required by each model, suggests that `FlamMap` is more suitable for planning contexts due to its ability to capture fine-scale fuel-topography interactions, while `PROPAGATOR` is better suited for operational scenarios thanks to its flexibility, fast implementation and computation, and capacity to incorporate time-varying wind.

The analysis benefited from detailed information provided by local managers directly involved in the events, which helped reconstruct the actual fire dynamics and evaluate model outputs against real-world cases. These findings emphasize the critical role of accurate wind and fuel data, as well as the need for a structured database of past wildfires in the Alpine context to improve model calibration and reliability. Overall, this study provides practical insights to support stakeholders in selecting appropriate modeling tools across different phases of wildfire risk management and underscores the need for further refinement of simulation systems tailored to complex mountain landscapes.

1. Introduction

Climate change is increasing the potential for wildfire occurrence due to more frequent and intense droughts, combined with extended periods of unfavorable fire weather conditions that reduce vegetation moisture and favor wildfire occurrence and spread (Popescu et al., 2022; Jones et al., 2024).

These factors, combined with the socio-economic changes, land abandonment and land use changes we have been witnessing in recent decades may exacerbate the fire regime in areas already heavily affected by the phenomenon, such as Italy (Mantero et al., 2020; Ganteaume et al., 2021). Italy's wildfire susceptibility is rooted in a combination of climatic, ecological, and anthropogenic factors, and the diversification of these components through the national territory (Elia et al., 2022; Kirschner et al., 2024). The Mediterranean climate that characterizes southern Italy and coastal areas, marked by wet springs that increase fuel availability, followed by prolonged summer droughts, frequent heatwaves, and strong winds, creates highly flammable conditions during the summer season. On the other hand, the Temperate climate of the norther Italy, Alpine Arc and Apennines mountains, is characterized by windy and dry fuel conditions especially in winter and spring seasons, favoring there wildfire activity during this period (Blasi et al., 2019; Trucchia et al., 2022). The country's complex topography and fuel-rich vegetation further exacerbate fire spread and intensity. Moreover, human activities such as land abandonment, agricultural burning, and accidental or intentional ignitions significantly contribute to fire occurrence (Ascoli et al., 2024). Climate change is intensifying

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these dynamics worldwide, as well as in Italy, by lengthening the fire season and increasing the frequency of extreme weather events (Duane et al., 2021; Richardson et al., 2022), however with marked regional contrasts around the World (Doerr and Santín, 2016). In 2023 alone, Italy experienced 4,265 forest fires, a 34.7% decrease compared to 2022, yet the average area burned per fire nearly doubled, rising from 10.9 to 20.8 hectares, resulting in a net increase in total burned area (San-Miguel-Ayanz et al., 2018; Berardi et al., 2025).

The Alpine region is particularly vulnerable to the increasing wildfire risk due to rising temperatures and shifting precipitation patterns, indicating a substantial worsening of fire weather conditions (Cane et al., 2013; Hetzer et al., 2024). Moreover, forest stands in the Alpine range, such as temperate conifer forests (Arrogante-Funes et al., 2024), and forests dominated by thin-barked species like *Fagus sylvatica* and *Abies alba*, exhibit high mortality rates after wildfires (Maringer et al., 2016; Dupire et al., 2019), as these species have not evolved within a fire-adapted regime. This peculiar set of conditions have contributed to challenge wildfire suppression efforts and wildfire risk management systems. The increase intensity of wildfire events across the Alpine Arc, together with the high sensitivity of its forest species, calls for a new approach centered on mitigating impacts through preventive management (Seidl et al., 2017), and for dedicated research on this region (Müller et al., 2020).

The implementation of preventive measures requires increased efforts from the scientific community to adopt integrated fire management and innovative modeling strategies to support stakeholders involved in wildfire risk management (Chuvieco et al., 2023). Wildfire modeling plays a crucial role in understanding the complex relationships between fire occurrence, driving factors, and impacts, addressing the prevention, suppression, and recovery phases. This comprehensive framework can be applied to decision-making processes at various levels, ranging from landscape management to household protection and evacuation planning (Oliveira et al., 2021).

In this framework, two widely used approaches to wildfire modeling are: (i) empirical and semi-empirical models, which rely on statistically derived laws of fire propagation (Rothermel, 1972); (ii) stochastic grid-based models, where evolving quantities are typically described through spatial and temporal discretization by implementing local interactions. These models simulate the propagation of the fire front from one cell to its neighboring cells by applying detailed localized evolution rules that incorporate either the underlying physics at the desired resolution or semi-empirical rules of fire propagation (Alexandridis et al., 2011). See Sullivan (2009b,a) for a more comprehensive description of wildfire modeling approaches.

Although it is widely acknowledged that none of these modeling frameworks can be universally deemed the "best" for wildfire risk management, stakeholders often require guidelines to select the most appropriate approach for each specific task, as each method has its own strengths and limitations. No model is perfect but some models are useful, depending on the objective and constraints required.

Empirical models are relatively straightforward to implement and require minimal computational resources. They can approximate fire-spread dynamics within known constraints due to their simplified structure. However, since most of these models are derived from controlled laboratory experiments, their reliability in predicting wildfire behavior under different conditions and within complex landscapes (e.g., Alpine environments) remains limited (Sullivan, 2009a).

In the European and Italian context, `FlamMap` — a semi-empirical model — is widely used for wildfire risk assessment (Mitsopoulos et al., 2015; Yavuz et al., 2018; Regione Piemonte, 2021; Mauri et al., 2024). Developed by the *United States Department of Agriculture Forest Service* (USDA) (Finney, 2006), this software enables researchers and fire management professionals to simulate fire spread, intensity, and potential impacts across a range of environmental scenarios, thereby supporting wildfire preparedness and response planning.

Grid-based stochastic models are lightweight, versatile, and easily integrable with existing databases. With some preliminary setup, they can effectively account for spatial and temporal heterogeneity, such as variations in vegetation types, topography, and meteorological conditions. Furthermore, these models can be seamlessly coupled with Geographic Information Systems (GIS) and can use (potentially real-time) meteorological data as inputs (Alexandridis et al., 2011). Among these models, stochastic Cellular Automata (CA) models for wildfire simulations are some of the most well-known examples. CA models discretize spatial interactions using square or hexagonal grids (Encinas et al., 2007), with fire spreading from burning cells to neighboring ones through probabilistic rules. Although CA models may simplify the underlying physical processes, their modular nature allows them to achieve the desired levels of complexity and accuracy.

One such example used in Italy is `PROPAGATOR`, a semi-empirical stochastic CA model (Trucchia et al., 2020), developed in 2008 by the *CIMA Research Foundation*[1] for the Italian Civil Protection Department. The primary objective of the model is to rapidly simulate possible wildfire evolution, providing scenarios during the response phase to ongoing emergencies (Perello et al., 2024). In Italy, the model is currently available to the Italian Civil Protection Department through a user-friendly web application in MyDewetra platform (Italian Civil Protection Department and CI.M.A. Research Foundation, 2014) . Since its initial release in 2008, the model has been applied and tested in numerous national and European projects (Consortium, 2020, 2019; Lantero et al., 2022). Stakeholder feedback has been crucial in enhancing its performance and capabilities. Despite these advancements, `PROPAGATOR` has predominantly been utilized as a rapid-response tool for emergencies (Trucchia et al., 2024), with limited application in long-term forecasting and planning (Perello et al., 2024).

Both models serve as references used in Italy in different contexts and for different purposes, also reflecting the diversity of expertise and institutions involved in the field of wildland fire management in the country (Sapountzaki et al., 2011; Kirschner et al., 2024; Verde et al., 2024). `FlamMap` is actively used especially during wildfire risk assessment and land-use planning phases, particularly for drafting the regional plans for wildfire risk management (i.e., *piani AIB*, Kirschner et al., 2024). The `PROPAGATOR` model, on the other hand, is used within the context of emergency response and civil protection, as it is made available to the Italian Civil Protection Department.

Given the prominent role of these wildfire modeling tools in the Italian fire management system, at both regional and national levels, and the need for targeted research on wildfire risk management in Alpine environments, this study investigates the potential applicability and reliability of the `PROPAGATOR` and `FlamMap` models within the Alpine Arc. Although some studies are emerging on the use of `FlamMap` in the Alpine context for wildfire risk mapping are emerging (Mauri et al., 2025, 2024), to be the best of our knowledge, no comparison studies between these two models in the Alpine region is present, which can support in a more accurate decision on the use of the two models for different application in wildfire management. The aim is to understand whether these models are suitable for supporting the distinct phases of wildfire risk management in this context – namely planning, prevention, and emergency response – in terms of required data and operational applicability; and what limitations may arise from the models themselves or from the availability and characteristics of the input data needed for their effective operational use in the Alpine context. To explore these questions, we applied both models to three

[1] CIMA Research Foundation is a research organization dedicated to disaster risk mitigation, with a particular focus on climate change-related hazards, including floods, wildfires, droughts, and the loss of terrestrial and marine biodiversity. Additionally, it serves as an operational structure within the National Civil Protection Service, as established by Italian Legislative Decree 1/2018 (Civil Protection Code).

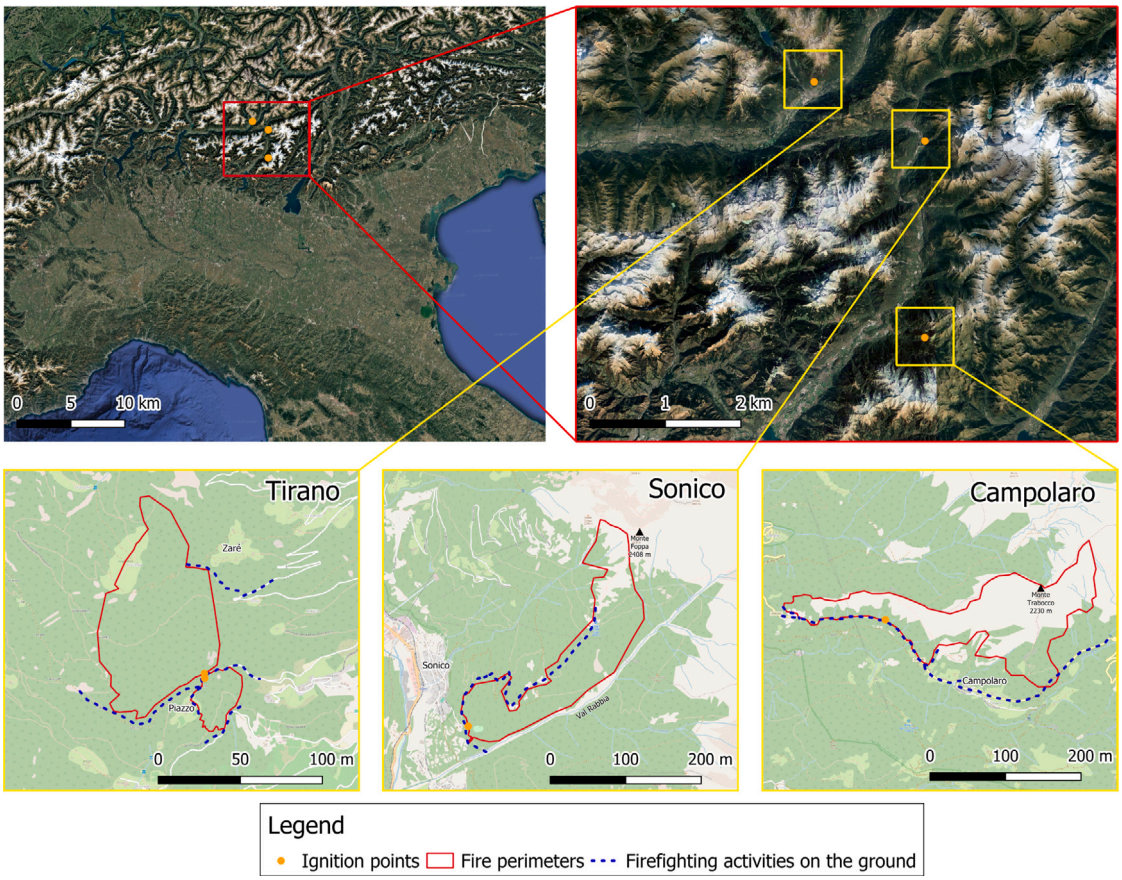


Fig. 1. Geographical context of the three case studies and overview of the main territorial information.

case studies corresponding to wildfires that occurred in Lombardia region (Italy) in recent years, each under different meteorological and propagation conditions: (i) fuel and morphology-driven fire, (ii) wind-driven fire, and (iii) fire with complex wind and spread dynamics. By examining the required input data, potential outputs, and model behavior under different propagation regimes, the results aim to support stakeholders in selecting the most appropriate model for the various phases of wildfire management – ranging from planning and prevention to decision-making during suppression activities – and to highlight potential areas for improvement to enhance their effective use in Alpine environments.

The paper is structured as follows: the next section describes the case studies, followed by a brief presentation of the models and the simulation settings adopted. Subsequently, the results are presented and discussed in the dedicated sections. The paper concludes by summarizing the main findings related to the use of the two models in wildfire risk management, with particular attention to the Alpine context.

2. Material and methods

2.1. Case studies

This section describes the three case studies selected for analyzing the behavior of the two models. The events were chosen among the wildfires occurred in Lombardia region (in northern Italy) between 2017 and 2022, based on relevance, data availability, and variability in environmental and operational conditions.

Relevance was assessed by considering the significance of each wildfire in terms of size – obtained by official burned area datasets – impact on local communities, and operational complexity. Starting from the regional wildfire dataset for Lombardia (2007–2022), only events larger than 50 hectares were considered. Among these, priority was given to wildfires that occurred in relatively recent years, to ensure the availability of reliable input data, and that presented sufficient documentation regarding fire progression and suppression activities. The information regarding impact on local communities and operational complexity, and the different suppression activities implemented, were collected through interviews conducted with local government officials and emergency management personnel, who supported in the selection of the events. The data availability on different wildfire events imposed an important constraint in events' selection, given the lack of systematic data collection at national level — some comments on this will be considered in the Discussion section. The final selection included three wildfires that differed in size, terrain, vegetation, and firefighting dynamics, allowing for a representative comparison across diverse conditions.

The selected events represent three different wildfire scenarios, with varying propagation conditions and increasing levels of complexity. The *Case Study A (Tirano Fire)* represents wildfire driven by fuel and topography; the *Case Study B (Campolaro Fire)* represent a wind driven wildfire; and the *Case Study C (Sonico Fire)* represents complex wind and spread patterns, with different phases of spreading and reignition.

In Fig. 1, the geographical context of the three case studies is presented. The fire regime of the region is characterized by wildfires mainly occurring in the Alpine area during the winter and spring seasons, typically driven by morphology and wind conditions, as well as particularly dry fuel conditions following the end of winter season. Regarding wind conditions, it is typical for *Foehn* wind conditions to develop in the area due to the presence of Alpine Arc, a warm, dry, downslope wind descending from the mountains toward the valleys, which favor wildfire occurrence and spread.

Table 1
Comparative summary of the three wildfires.

Case Studies	Tirano Fire	Campolaro Fire	Sonico Fire
Period	March 11–15, 2022	January 4–6, 2017	March 23–29, 2022
Affected area	70 ha	260 ha	211 ha
Weather conditions	Moderate wind and dry fuels	Strong wind (Foehn), dry fuels	Moderate wind with complex topographic-driven pattern, dry fuels
Propagation dynamics	Topographic fuel-driven fire	Wind-driven fire	Complex spread dynamics, fire re-ignition and particular convective trigger on the top
Fire-fight actions	Helicopter activity on flanks, ground teams concentrate on the right side in protection of houses in the hamlet of Zare. Containment activities on the right low to the burned area. Containment on the road in the lower part of the wildfire.	Important activity on the right side downstream near the township. Tail containment activity on the left. On the second day, activity of aerial and ground assets on the left side to slow the propagation.	Important activity on the left side to contain the spread with ground squad. Activity on the ridge of the right valley (ground squads) and helicopter activity to slow spread.

To collect information for this study, experts who were involved in the wildfires during or after the events were consulted. Directors of firefighting operations for the *Comunità Montana della Val Camonica* and from the Agricultural and Forestry Sector of the *Comunità Montana della Valtellina di Tirano*[2] provided insights into the Sonico and Campolaro wildfires, and Tirano wildfire, respectively. These professionals provided valuable details about the firefighting operations and the evolution of the wildfires, as well as important information about the local weather conditions during the events. The summary of the three events is reported in Table 1.

2.1.1. Case study A: Tirano fire

The first case study is the wildfire that occurred in Tirano municipality (Sondrio province, Lombardia region, Italy) on March 11, 2022, on the southern side of *Monte Masuccio*, facing Tirano. The fire started in the late afternoon near the houses in Piazzo (see Fig. 1), spreading slowly below the road for approximately 24 h, with firefighting efforts focused on the area below the road and near the houses. Moderate to low wind characterized the event, however dry fuels conditions typically observed in the region in the late winter period were present, with availability of fuel for spreading. On the morning of March 13, the fire reignited above the road, spreading faster northward, driven by the available fuels. Near the village of Zare, firefighting were carried out to protect the houses with ground crews. Helicopters were used on the left and right flanks to contain the fire's spread through aerial water drops.

2.1.2. Case study B: Campolaro fire

The second case study is the wildfire occurred in Campolaro municipality (Brescia province, Lombardia region, Italy) on January 4, 2017. The wildfire was located on the southern side of *Monte Trabucco*, in a lateral valley of *Val Camonica*, oriented east–west. The day was characterized by a strong, dry and hot wind from north (Foehn wind). The configuration of the valley, oriented from east to west, caused the channeling of the wind coming from the north along the Camonica Valley, leading to a significant change in direction. Based on information gathered from local experts, the wind in Campolaro during the event was blowing from the west along the valley's axis. The fire started in the afternoon above the road leading to Campolaro, spreading uphill and to the east, driven by the wind. Significant firefighting efforts were carried out along the road and near houses on the right flank, while containment actions were focused on the left flank of the fire's tail. On the second day, both aerial and ground resources were deployed on the left side to slow the fire's propagation.

2.1.3. Case study C: Sonico fire

The third case study is the wildfire that occurred in Sonico municipality (Brescia province, Lombardia region, Italy) on March 23, 2022. Located on the right side of *Val Camonica* above the town, it spread in the lateral valley of *Val Rabbia*, reaching almost the top of *Monte Foppa*. The wind conditions during the event were typical for the area in spring, following the usual wind patterns shaped by the terrain, namely: morning uphill wind and afternoon downhill. The ignition point was located south of Sonico at the edge of the forest. The fire began spreading uphill, following the slope and the fuels characteristics. The spread was halted on the left flank by ground crews. At the top of Val Rabbia, a line of firefighters was established to contain the fire along the ridge. However, on the March 27th, after an apparently quiet phase, the fire reignited, spreading throughout the valley, reaching the ridge, and continuing into the upper grasslands with a particular convective trigger on the top which allowed the fire to reach the opposite slope.

2.2. Propagation models

In the following paragraphs, a brief description of the two models used in this study is presented.

2.2.1. FlamMap model

`FlamMap` 6.2 (Finney, 2006) is a fire behavior modeling software that simulates potential fire spread and intensity under constant environmental conditions. It is available as a free desktop application (https://research.fs.usda.gov/firelab/projects/flammap). `FlamMap` uses Rothermel's surface fire rate of spread model (Rothermel, 1972) along with a two-dimensional Minimum Travel Time (MTT) algorithm (Finney, 2002) to predict fire growth patterns and delineation of wildfire spread. Unlike dynamic fire simulators, `FlamMap` does not model temporal variations in weather or suppression actions, making it ideal for assessing fire behavior potential across landscapes, useful for risk mapping and risk mitigation planning (Mauri et al., 2024).

[2] A *Comunità Montana* is a form of local authority established under Italian national law (Law No. 1102 of December 3 1971, and Legislative Decree No. 267/2000). It is composed of a group of neighboring municipalities located in mountainous or partially mountainous areas, with the main purpose of promoting economic, social, and environmental development in the area.

Table 2
Simulation settings of each model.

	FlamMap		PROPAGATOR	
Landscape inputs	Elevation Slope Aspect Canopy cover Fuel Models (Scott, 2005)		Elevation Fuel Models (PROPAGATOR)	
Weather inputs	Wind Fuel moisture		Wind Fuel moisture	
Outputs	Arrival time Major paths Rate of spread Fireline intensity		Fire probability Isochrones Rate of spread Fireline intensity	
Settings	Resolution	10 m	Resolution	20 m
	Interval for Minimum Travel Paths	100	Spotting	no
	Spot probability	0.001		
	Spot delay	0		
	Lateral search depth	6		
	Vertical search depth	4		

The software requires the creation of a Landscape file, containing all the vegetational and morphological information. The landscape information required by the software includes elevation, slope, aspect, canopy cover and fuel model maps - the latter referred to either Scott (2005) or Anderson (1982) fuel classifications. Regarding weather information, wind (speed and direction) and fuel moisture conditions must be specified, which are considered fixed during the whole simulation.

FlamMap provides static maps of potential fire behavior based on these fixed environmental conditions. The main outputs include *Rate of Spread*, *Fireline Intensity*, and *Flame Length*, offering insights into fire dynamics across different scenarios based on inputs provided. Additionally, the MTT algorithm within FlamMap simulates the minimum travel path of fire, calculating its direction and speed of propagation over a heterogeneous landscape, and producing then fire front arrival times and major propagation pathways, essential for planning suppression strategies and risk management. The software can be either used for simulating single events (as done in this study) or for planning by simulating multiple artificial events through the landscapes. In the latter case, the software produces also fire probability maps, often used for landscape planning (Mollocana-Lara et al., 2025).

2.2.2. PROPAGATOR model

PROPAGATOR is a wildfire simulation model based on a stochastic Cellular Automata framework (Trucchia et al., 2020, 2024; Perello et al., 2024). The model was developed to provide a decision-support tool during emergency situations, with particular focus on Civil Protection. For this reason, simulation rapidity and simple input data are the core aspects of the model. The model is available as a python open source code on GitHub (https://github.com/CIMAFoundation/propagator_sim) as well as provided though a user-friendly web-GIS application (i.e. MyDewetra platform, Italian Civil Protection Department and CI.M.A. Research Foundation, 2014) within specific collaboration agreements with the developer entity.

The model takes elevation and fuel type as input, using a custom simplified fuel classification specific to the model, namely: grasslands, shrubs, broadleaves, fire-prone conifers, agro-forestry areas, non fire-prone forests, and non-vegetated areas. Fuel information is typically derived from land-use maps, representing a coarse classification of fuel types for rapid simulations in the emergency phase. Additionally, the model incorporates wind and fuel moisture data. Dynamic inputs can be provided at an hourly or even minute-by-minute resolution during the simulation, along with firefighting activities, supporting the dynamic description of the event and the planning of suppression operations during the emergency phase. Specifically, the model includes four possible suppression actions: *Canadairs*, helicopters, waterlines, and heavy actions (i.e., the extensive use of suppression actions, machinery for fuel removal, or similar firefighting measures that create a barrier for the fire propagation). A *spotting* model can also be activated (López-De-Castro et al., 2024) i.e., to include the transport of burning pieces of material by wind, which can ignite new fires at a distance from the main fire.

The input data drives the probabilistic propagation of the fire between grid cells following the ignition(s). As a stochastic framework, the model runs an ensemble of multiple simulations, providing outputs at each simulation time step (e.g., every hour), including: the fire probability map (i.e., the probability of each cell to be affected by the wildfire within that specific time step); isochrones of the most probable burned areas at different thresholds - typically greater than 50% or 75% probability; rate of spread and fireline intensity maps.

More details on the model and equations can be found in Trucchia et al. (2020).

2.3. Models set up

After reconstructing the dynamics of the case studies, the data needed for the simulation of the events were collected, and the simulations were configured to reflect the progression of the events. The summary of the simulations settings is reported in Table 2.

2.3.1. Models input data

In the following paragraphs, the description of the input data for simulations is presented.

Landscape inputs. Different static inputs are needed in both the models to represent the landscape information.

The topography information represent a common static input of the two models: while PROPAGATOR uses only an elevation file, other derived topographic information (e.g., slope and aspect) are required by FlamMap. For the Elevation layer of the two models, the Tinitaly DEM (Tarquini et al., 2007) was used, while the Slope and Aspect layer needed by FlamMap were calculated on QGIS using GDAL algorithms (QGIS version 3.34.9, 2024) starting from the DEM data.

Canopy cover and fuel model layers are needed to FlamMap. For the Canopy cover layer the Tree cover Density 2018 from Copernicus (Copernicus Land Monitoring Service, 2019) at 10-meter resolution was used. The last layer needed by FlamMap, i.e. Fuel Model, was created starting from the Fuel Type association from *annex 8* of the *Piano Antincendio Boschivo*[3] of Lombardia region (i.e., the regional

[3] Regions in Italy coordinate their own wildfire management system, providing a regional plan which serve as framework for the wildfire-related activities at the regional level.

Table 3
Summary of meteorological data sources and periods for each case study, and of the moisture scenarios used for the two models. The RISICO fuel moisture presented here is intended as the mean value obtained during the event.

Case study	Tirano	Campolaro	Sonico
ARPA meteorological station	Tirano – Eliporto	Breno – Lago della Vacca	Edolo – Centrale ENEL
Coordinates (latitude–longitude)	46.20° N – 10.21° E	45.95° N – 10.45° E	46.17° N – 10.37° E
Data Period	11 March–15 March	4 January – 6 January	23 March–29 March
Moisture from RISICO	15.0%	12.2%	12.7%
Live Fuel Moisture scenario (FlamMap)	L2	L2	L2
Dead Fuel Moisture scenario (FlamMap)	D4	D3	D3

wildfire risk management plan) (Lombardia, 2024), which has been used to join fuel type with forest types and land cover to create a map for each case study. After that, at each fuel type was associated with a Standard Fuel Model (Scott, 2005) required by FlamMap.

For what regards the fuel information for PROPAGATOR, CORINE 2018 land cover map was used (European Environment Agency, 2020), as considered for the operational implementation of the model within the Italian Civil Protection system, associating the different classes to the customed fuel types described before. The different landscape inputs needed by the models are listed in Table 2.

Weather conditions. The weather information required by both the models includes wind speed and direction, as well as fuel moisture content.

Wind-related data were obtained from the ARPA[4] meteorological station closest to each event (see Table 3).

Considering that in FlamMap all fire behavior calculations assume constant wind speed and wind direction over time (Finney, 2006), the mean wind speed and predominant direction were calculated based on the hourly average data for the entire duration of each event. The wind information have been used in FlamMap as input for the WindNinja option within the model. This tool integrates wind speed and direction with the landscape's topography to generate wind grids and vectors, thereby providing spatialized wind information. For PROPAGATOR, given its capability to accept time-dependent inputs, the original observed wind data were used - however, no spatialization of the wind is considered in this model.

Due to the specific morphology of the Campolaro Valley and the distance between the ARPA weather station and the fire location, the recorded wind direction was adjusted to better reflect local conditions of the Campolaro Fire event. Based on observations from the incident commander on-site, it was assumed that the wind followed the natural axis of the valley, as this was deemed more representative of the actual fire behavior. This wind direction modification has been implemented for both the models. In addition, for this event two wind speed scenarios were tested for FlamMap: one based on the average speed excluding minimum values, and another using the maximum recorded speed. The two scenarios have been tested to reflect the strong wind conditions observed during the event. For PROPAGATOR, given the possibility of including hourly wind variations and then sudden wind changes, the hourly recorded wind speed has been used.

It is important to note that the available wind data for all the wildfires were already provided as hourly averages, meaning that short-term strong gusts, which could significantly influence fire propagation, were not captured.

Since no sensors were available to directly measure fuel moisture content, information from the RISICO model was used instead (Perello et al., 2025).

Regarding fuel moisture data, the FlamMap model requires a fixed moisture content for each fuel size class (1-hr, 10-hr, 100-hr, live herbaceous, and live woody). Since the RISICO outputs are not categorized by fuel size, this information was used to select the appropriate moisture scenario from Scott and Burgan's Standard Fire Behavior Fuel Models (Scott, 2005) - it is described in Table 3. For PROPAGATOR, a single proxy of fuel moisture is required - the hourly fuel moisture time series directly derived from the RISICO model were used.

2.3.2. Simulations settings

To finalize the input for both models, the simulation parameters were configured. The ignition points and the Time Simulations were set based on the wildfire events dataset and expert input (see Table 4). For what regards Tirano fire, the event was split into two parts, following the first spreading below the road (1440 min) and then the new ignition in the upper part (after the first part, for 2160 min). For what regards Sonico fire, the same approach was adopted, splitting the fire into a first part (4320 min) and then the new ignition which accounted for much of the burned area (after the first part, for 2880 min). In this case, the new ignition was set starting from the first fire burned area. The need to split these two events was driven by the need to reflect the two expansion phases that were observed during both the events, and to better reflect the observed behavior within the simulations. Regarding Campolaro Fire, a unique ignition was set following the observed dynamics of the event.

For what regards FlamMap simulation setting, the Resolution of calculations (i.e., 10 m) and the Interval for Minimum Travel Paths were adjusted to achieve a balance between computational speed and result accuracy. Other parameters, such as Spot Probability, Spotting Delay, Lateral Search Depth, and Vertical Search Depth, were kept at the software's default settings. For PROPAGATOR, no further parameters setting are needed apart of the spotting option — no spotting model was adopted given the missing information on spotting dynamics. With respect to FlamMap spatial resolution, the PROPAGATOR input and output have been provided at 20 m resolution following the operational implementation of the system. The different simulation settings for the two models are reported in Table 2.

For both the models, the firefighting activity obtained by the insights of the personnel directly involved in the events, was used setting them barriers. In particular, to make the results comparable and following the FlamMap settings, the heavy action option was adopted for PROPAGATOR without any further distinction. The areas where firefighting activity has been considered in simulations can be observed in Fig. 1.

2.3.3. Models outputs data

To compare the simulations from the two models, the outputs were chosen to describe fire behavior and be compatible between the models. Fireline Intensity and Rate of Spread, key fire behavior parameters, are common to both models and serve as a good basis for comparison. Arrival Time and Major Paths outputs from FlamMap were used to analyze the propagation patterns of each model and compare them with the actual fire perimeter recorded in the dataset. For PROPAGATOR, the fire probability obtained in the last time step, and the different isochrones obtained during the simulation time (with thresholds set at 50% probability), were used for the same purpose. The different outputs of the two models are reported in Table 2.

[4] ARPA (*Agenzia Regionale per la Protezione dell'Ambiente*) is Italy's network of regional environmental protection agencies. Each ARPA monitors air, water, soil, and environmental risks, supporting public authorities with data, assessments, and compliance checks. They operate regionally but are coordinated at the national level through the SNPA system (*Sistema Nationale Protezione Ambiente*).

Table 4
Ignition points and time simulation settings for each case study.

Case study	Tirano	Campolaro	Sonico
Ignition Point (latitude–longitude)	46.22° N - 10.20° E	45.92° N - 10.38° E	46.15° N - 10.38° E
Time Simulation (min)	1440/2160	1440	4320/2880

3. Results

As previously described, two simulations were carried out for each case study – one for each model – in order to compare their results and highlight any differences, using real wildfire data as a reference to assess the reliability of the simulations. The arrival time and major paths generated by FlamMap were compared with the isochrones and fire probability maps produced by PROPAGATOR, using the actual fire perimeter to verify whether the simulated fire progression aligns with the real fire spread. To evaluate how the models reconstruct the main descriptors of fire behavior, the rate of spread and fireline intensity produced by the two models were compared to identify any qualitative differences.

3.1. Case study A: Tirano fire

The outputs of the two simulations, the one below the road and the one spread uphill were visualized together in Fig. 2. The Arrival Time output of FlamMap shows a credible burned area, closely matching the actual fire perimeter (Fig. 2a) and the Major Paths generated by the model depict a plausible evolution of the fire. Given the stochastic nature of PROPAGATOR, the fire probability map has to be read together with isochrones, which highlight the areas with greater probability. The fire probability map from PROPAGATOR shows a broader potential fire spread extending well beyond the actual perimeter, particularly along the slope in a northerly direction, but with very low probability (Fig. 2b). The possibility of further spread in PROPAGATOR is mainly due to the continuity of fuel and the topography of the area, although the relatively high fuel moisture values depicted for this event limits the spreading probability to lower values. As result, the area with higher fire probability is limited to the area of the actual fire perimeter, aligning with the real scenario and with FlamMap. The isochrones, in fact, depict a fire spread more consistent with the actual event.

Both models identify limited fire spread during the initial phase of the event (i.e., the lower perimeter), followed by more extensive spread in the upper, second perimeter - it is consistent with what observed during the real event according to the information gathered on this event (see Material and Methods section). In the second phase, both models indicate the potential for a main propagation direction aligned with the local topography, consistent with the fire type (i.e., topographic wildfire).

The comparison of Fireline Intensity (FI) and Rate of Spread (ROS) reveals that the two models operate within completely different value ranges (see Fig. 2c,d and Fig. 2e,f respectively). FlamMap outputs show higher values of ROS and FI in the western part of the fire, along with some extreme hotspots in the central area. In contrast, the output from PROPAGATOR appears highly variable, with scattered values due to the cellular automata nature of the model, which operates cell-wise. An increase in ROS and FI is observed in the western part of the wildfire, as pointed out by FlamMap, and in the right side (see Fig. 2c and Fig. 2e, respectively). In both models, the first part of the wildfire (e.g., the lower part of the event) is characterized by low rate of spread and intensity.

3.2. Case study B: Campolaro fire

The Campolaro fire was characterized by strong wind conditions. The influence of wind in this case study is evident from the FlamMap simulations carried out using two different wind speed scenarios. In both cases, the simulations reproduce the main direction of fire spread; however, there is a significant difference in the extent of the burned area depending on the wind speed (compare Fig. 3a and Fig S.2 supplementary materials). For comparison with PROPAGATOR, the second scenario – with a wind speed of 36 km/h – was selected to be more coherent with the insights from local managers and with the final burned area. Both models significantly underestimate the westward propagation of the fire, rather advances eastward along the upper part of the slope. FlamMap fails to capture the fire's downslope spread, which is better represented by PROPAGATOR, although with lower probability (see Fig. 3b). Overall, PROPAGATOR more accurately reproduces the shape of the fire perimeter (Fig. 3b), while FlamMap provides more detailed information on the internal dynamics of fire spread within the burned area for what regards the other fire behavior descriptors.

As with the first case study, the fire behavior descriptors reveal differences in both scale and the type of information provided by the models. FlamMap highlights two areas with high values of Rate of Spread and Fireline Intensity, offering insights into the fire's spread dynamics (Fig. 3c and Fig. 3e). PROPAGATOR, on the other hand, shows high values during the initial phase of the fire, and in the middle area of the event (Fig. 3d and Fig. 3f). Additional high values appear in the lower section of the area, which is not considered in the FlamMap simulations given that the model does not simulate a fire spreading in this area.

3.3. Case study C: Sonico fire

The complexity of the Sonico fire – both in terms of wind conditions and fire spread dynamics – made this case the most challenging to analyze. The outputs from the two model simulations, carried out to capture the fire's re-ignition, were combined into a single output for comparison purposes. In the first FlamMap simulation (Fig. 4a), the fire spread across the slope, stopping only due to the barrier created by firefighting actions. The PROPAGATOR simulation (Fig. 4b) shows a similar spread pattern, though with a more limited area and more independent with respect to the application of firefighting actions. In contrast, in the second simulation, FlamMap displays slower fire propagation, following the upper part of the valley as the main direction. It underestimates the final fire perimeter and fails to capture the specific dynamics of the later phases of the event. PROPAGATOR, on the other hand, more accurately follows the actual fire perimeter, even overestimating the spread in the upper section. However, the final isochrones produced by PROPAGATOR highlighting the most probable area are very similar to those of FlamMap.

The comparison of Fireline Intensity and Rate of Spread shows similar characteristics to those observed in the other case study (see Figs. 4c,d and Figs. 4e,f). The FlamMap simulation displays high intensity and rapid expansion during the initial phase of the fire, while in the later phase, both indicators show lower values, except in a localized area. PROPAGATOR, in contrast, presents more uniform FI and ROS values across the entire perimeter, with higher values concentrated in the upper parts of the first and second phases of fire spreading.

4. Discussion

Unlike other comparative modeling studies (Khalaf et al., 2024; Mohammad Beyki et al., 2025), this work not only analyzes the differences between model outputs but also aims to identify the main limitations and shortcomings of the models when applied to the Italian Alpine context The valuable collaboration with local fire managers

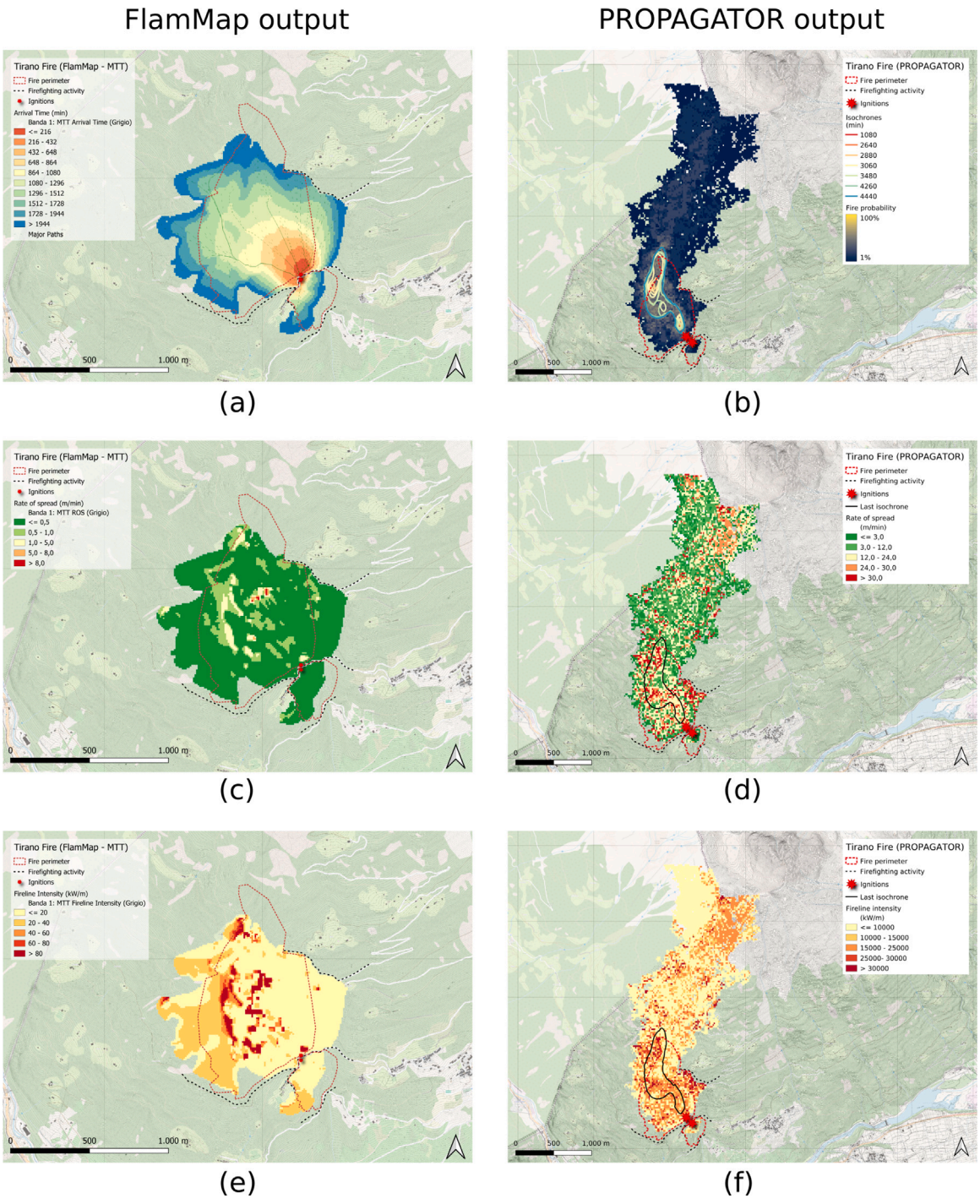


Fig. 2. Tirano Fire: (a) Arrival time and major paths from FlamMap; (b) Isochrones and fire probability from PROPAGATOR; (c) rate of spread from FlamMap; (d) rate of spread from PROPAGATOR; (e) fireline intensity from FlamMap; (f) fireline intensity from PROPAGATOR.

directly involved in the events enabled a more accurate reconstruction of wildfire dynamics, allowing for a grounded comparison of model outputs against real-world cases.

Our findings confirm that FlamMap and PROPAGATOR serve complementary roles within the wildfire risk management cycle. FlamMap, with its capacity to model fine-scale fuel-topography interactions, is particularly suited for strategic planning and prevention. In contrast, PROPAGATOR demonstrates greater operational flexibility, offering rapid simulations and the ability to incorporate dynamic inputs such as time-varying wind and suppression actions. This distinction is consistent with recent applications in Italy, including the use of PROPAGATOR for prescribed fire planning in Liguria (Perello et al., 2024), and the deployment of FlamMap to assess fire severity in Sardinia (Salis et al., 2016).

In terms of fire evolution and simulated burned area compared to the actual one, the two models perform reasonably well, although with some differences. In the first case study, the two models produce broadly similar results, with FlamMap more specific capturing certain behavioral and propagation dynamics. In the second case, PROPAGATOR provides a more accurate reconstruction of the fire, due to

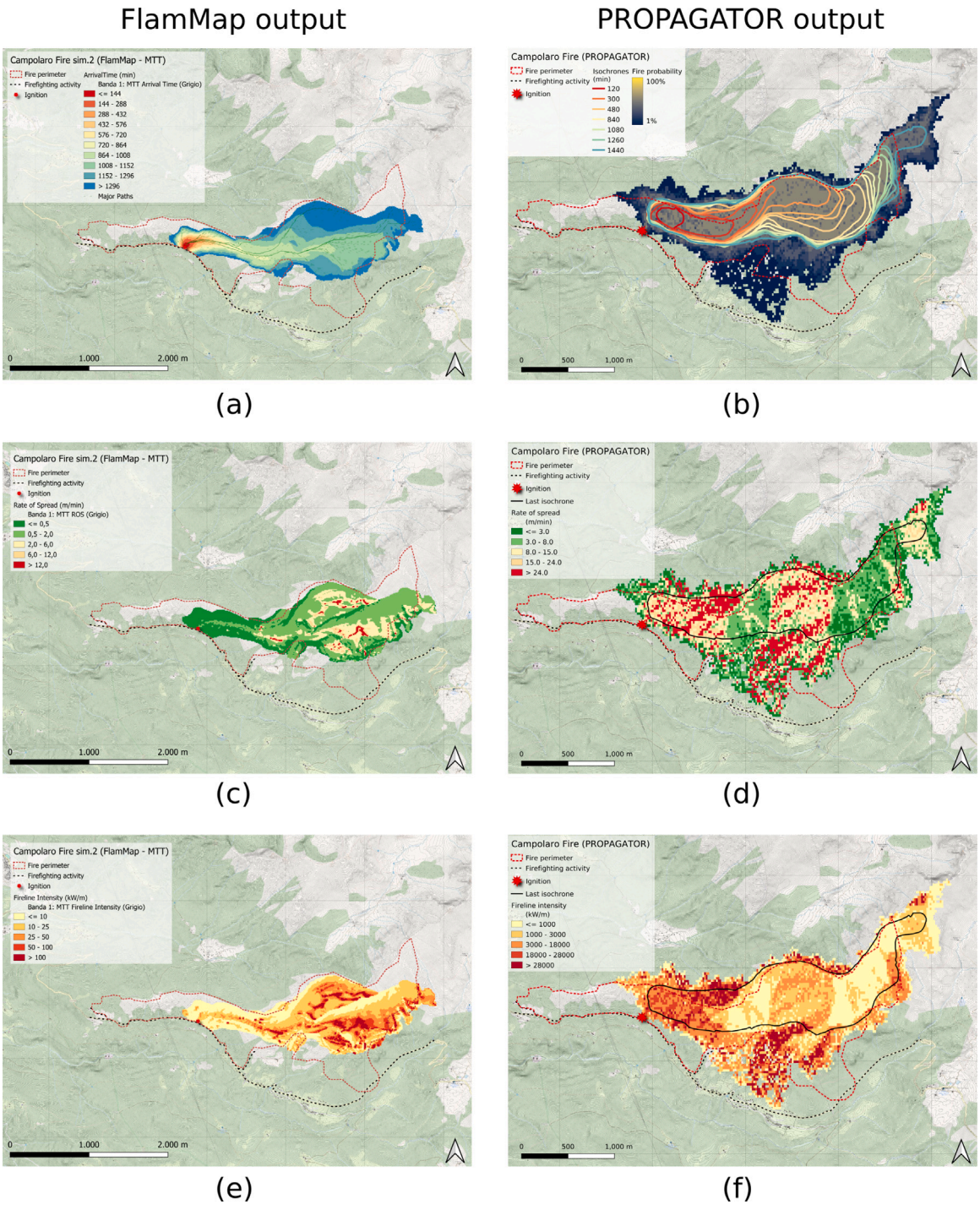


Fig. 3. Campolaro Fire: (a) Arrival time and major paths from FlamMap; (b) Isochrones and fire probability from PROPAGATOR; (c) rate of spread from FlamMap; (d) rate of spread from PROPAGATOR; (e) fireline intensity from FlamMap; (f) fireline intensity from PROPAGATOR.

its greater ability to simulate the significant impact of wind and its temporal variation. In the third and most complex case study, both PROPAGATOR and FlamMap struggle to reproduce in detail the complex dynamics that occurred during the final phase of the fire; however, PROPAGATOR appears to better capture the potential development of the fire front.

The fire behavior outputs (Rate of Spread and Fireline Intensity) reveal substantial differences between the two models.

PROPAGATOR results appear to be highly dependent on the fuel map adopted, with fire behavior values varying significantly according to land use changes. This is particularly evident in the Campolaro case study, where the change in fuel cover – from "forest" to "grassland" (see Figure S.1d in the Supplementary Material) – causes an abrupt change in fireline intensity and rate of spread.

PROPAGATOR assigns these two variables the role of proxies to guide firefighting actions, primarily aiming to highlight areas of general increase in rate of spread and fireline intensity for operational purposes, rather than providing a detailed representation of fire behavior. This approach emphasizes zones where a reduction in intensity could enhance suppression effectiveness or, on the other hand, increases in fireline intensity due to combination of land cover types, topography and weather conditions could complicate firefighting activities. These

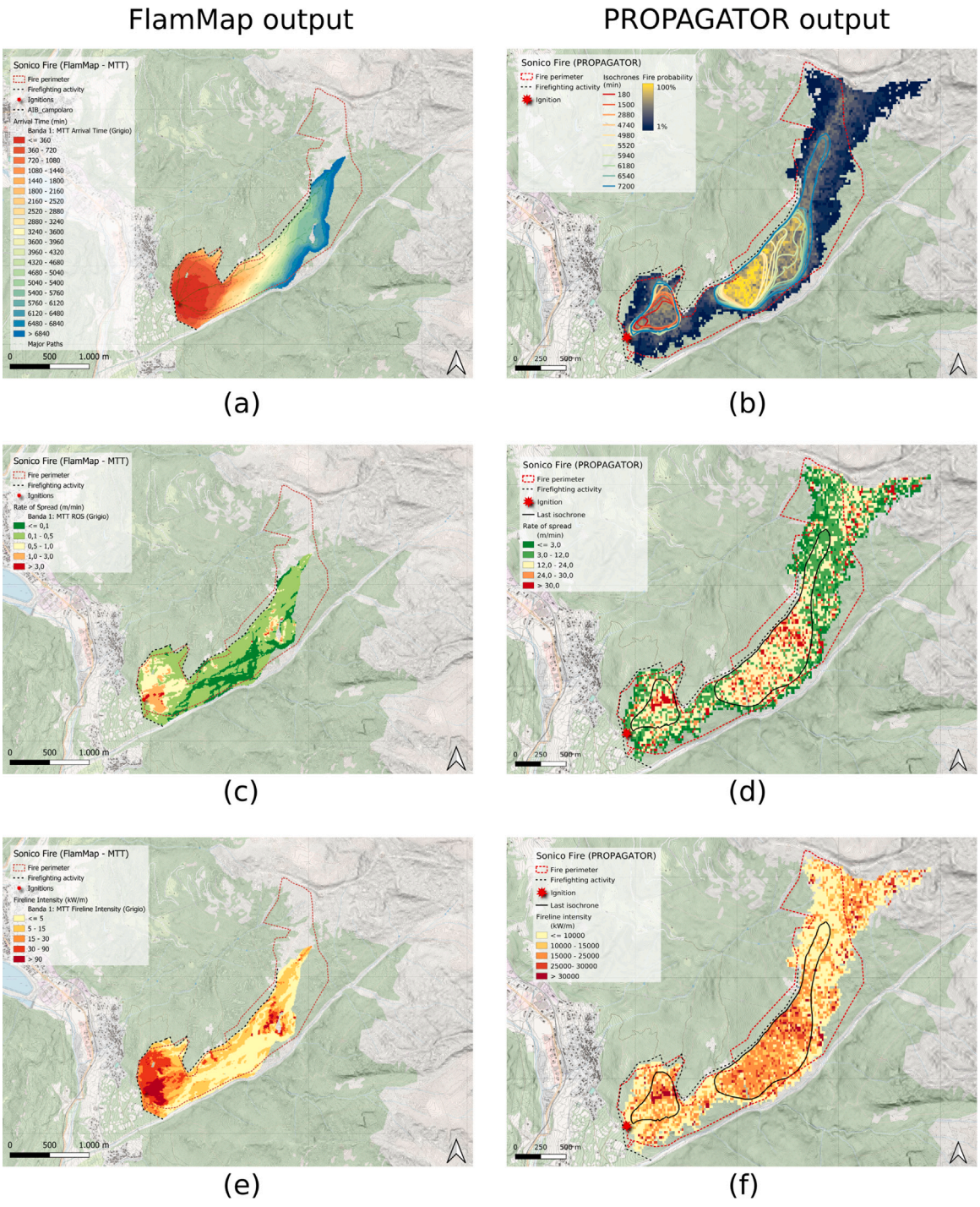


Fig. 4. Sonico Fire: (a) Arrival time and major paths from FlamMap; (b) Isochrones and fire probability from PROPAGATOR; (c) rate of spread from FlamMap; (d) rate of spread from PROPAGATOR; (e) fireline intensity from FlamMap; (f) fireline intensity from PROPAGATOR.

features combined with the model's ease and speed of use make it more suitable for emergency response.

In contrast, FlamMap, which is based on a different modeling approach that employs a more detailed fuel map derived from standard fire behavior fuel models (Rothermel, 1972; Finney, 2002; Trucchia et al., 2020), is better able to capture intra-fuel behavioral dynamics. These dynamics are more closely related to the interaction between fuel characteristics and topographic conditions. These differences in map availability and model usage make FlamMap more suitable for planning-oriented analyses, where the absence of time constraints in the simulation allows for the acquisition of detailed inputs such as the fuel map, and where detailed outputs linked to management prescriptions are essential.

The difference between the two models in handling wind dynamics is evident. PROPAGATOR supports temporal variability in wind speed and direction, allowing it to simulate dynamic changes that can result in substantial variations in fire behavior. This leads to a more accurate reconstruction of wildfire evolution under fluctuating wind conditions. Conversely, FlamMap incorporates spatial variability in wind as influenced by topography, enabling the model to represent topography-driven fire behavior with greater detail. This distinction becomes particularly relevant in scenarios characterized by pronounced

wind variability – such as Campolaro and Sonico Fires – where PROPAGATOR tends to perform more accurately. In contrast, under more stable and low wind conditions, as observed in Tirano Fire, FlamMap demonstrates better performances.

Wind is one of the main factors that influence the fire regime in the Alpine region (Zumbrunnen et al., 2009; Mauri et al., 2024). Our results, as expected, show that wind significantly affects the simulation outputs of both models; therefore, having access to accurate wind data is crucial for reliably modeling fire behavior, both in planning and operational contexts. The complex Alpine topography results in strong variations in wind behavior from one valley to another, as clearly observed in this study, making it difficult to obtain the information required for the simulations.

Similarly, fuel moisture conditions (both the live canopy component and the dead material such as litter) greatly influence fire behavior and the likelihood of wildfires, especially during particularly dry periods, such as following the winter season in the Alpine region under analysis. This aspect revealed for example in Tirano case study, where the effect of relatively high fuel moisture conditions affected the spreading probability simulated in PROPAGATOR. In our simulations, we used a fuel moisture model to approximate these conditions, assigning scenarios in FlamMap and using fuel moisture time series in PROPAGATOR. However, obtaining this information in a more detailed and spatially distributed manner would allow for better calibration and configuration of the simulation models, thereby developing more reliable scenarios for wildfire risk management.

The Alpine context is also characterized by a high degree of landscape heterogeneity, which results in markedly different fire dynamics depending on the vegetation cover and its conditions. Updating and producing maps that include essential information for simulations – such as fuel models – would not only accelerate the preparation of simulations either during planning or emergency phases, but also significantly improve their accuracy.

In all three case studies, implementing firefighting activities was challenging due to difficulties in obtaining accurate information. The impact of these activities is evident in the Sonico and Tirano case studies, where the barriers used to simulate ground operations altered fire spread. Firefighting efforts carried out by helicopters were not considered in the simulation due to the FlamMap model's inability to effectively represent such interventions. The differences observed in the lateral section of the Tirano fire (see Fig. 2a) are likely attributable to aerial suppression activities. The capability of PROPAGATOR to simulate a broader range of firefighting operations confirms its predominant role in emergency response and underscores the importance of accurately recording suppression efforts.

Despite the limitations described above and possibilities of refinements, the two models have proven to be effective tools for simulating wildfires in the Alpine context, each with a primary role in different phases of the wildfire risk management cycle. However, during the preparation and simulation of these case studies, some critical issues emerged, mainly related to the availability of essential data for analysis, on which it is important to focus future efforts. Improving the monitoring and mapping of wind, fuel moisture, and vegetation is essential to increase the accuracy and reliability of wildfire simulations (Rios et al., 2014). This includes developing high-resolution meteorological forecasting systems and producing updated, spatially explicit fuel maps tailored to Alpine conditions. Building structured databases of past wildfire events and suppression activities would support better model calibration and validation. Such resources would enable more robust scenario development for both planning and emergency response and facilitate the refinement of empirical and semi-empirical simulation tools.

5. Conclusions

The Alpine region represents a vulnerable context in light of the projected increase in wildfire activity under future socio-economic and climate change scenarios. Dedicated studies help to highlight the unique features and challenges of wildfire research in this environment.

It is essential to connect such studies to actionable insights for stakeholders, enabling informed wildfire risk management and enhancing the impact of scientific research. A crucial aspect in this regard is wildfire modeling and the evaluation of simulation tools tailored to the Alpine context.

This study examined two simulation models widely used in Italy — FlamMap and PROPAGATOR — to simulate three significant wildfire events in the Alpine area of the Lombardia Region. The analysis show that both models have proven to be effective tools in this setting, with their respective strengths and limitations identified in relation to different phases of the wildfire risk management cycle. FlamMap is particularly suited for planning and prevention due his fine-scale fuel-topography interactions although PROPAGATOR is better suited for operational scenarios thanks to its flexibility, fast implementation and capacity to incorporate time-varying variables.

The study highlights several critical challenges that must be addressed to improve wildfire modeling in the Alps, particularly those related to the availability and accuracy of input data. In this complex environment, detailed wind dynamics, fuel moisture data, vegetation information, and records of firefighting operations are often scarce or spatially inconsistent, which limits the capacity to parameterize and validate models effectively. Addressing these issues requires not only expanding and densifying local monitoring networks but also improving the fine-scale modeling of meteorological and fuel-related variables, together with the development of a comprehensive, standardized database of past wildfire events to support model calibration and validation.

Refining simulation tools adapted to the Alpine context would enhance preparedness for future events in the region. Furthermore, studies like the one proposed in this paper can provide stakeholders with valuable guidance on selecting appropriate tools for the various phases of the wildfire risk management cycle.

CRediT authorship contribution statement

Luca Malanchini: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Nicolò Perello:** Writing – review & editing, Visualization, Software, Methodology, Formal analysis, Data curation. **Andrea Trucchia:** Supervision, Conceptualization. **Lorenzo M.W. Rossi:** Writing – review & editing, Supervision, Conceptualization. **Paolo Fiorucci:** Writing – review & editing, Supervision. **Giorgio Vacchiano:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.envc.2025.101357.

Data availability

Data will be made available on request.

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